JAVA PROJECT REPORT

Title: Student Task Manager

Prepared by**: KHUSHI GAUTAM**

Registration Number: **24BAI10126**

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Introduction

Managing academic work has become increasingly challenging for students due to multiple assignments, deadlines, and daily responsibilities. Many students rely on memory or unorganized notes, which often leads to forgetting important tasks. This creates unnecessary stress and affects productivity.

To address this issue, a simple and effective solution is required that can help students keep track of their tasks in an organized way. The Student Task Manager is developed as a console-based Java application that allows users to manage their tasks efficiently.

This project focuses on providing a minimal yet functional system where users can add new tasks, view existing ones, and remove completed tasks. It is designed to be easy to use and does not require any advanced technical knowledge. Additionally, this project demonstrates how basic programming concepts can be applied to solve real-life problems.

Problem Statement

In the academic life of students, managing multiple tasks such as assignments, homework, project submissions, and exam preparation can be challenging. Students often rely on memory or unorganized methods like rough notes, which are not reliable for keeping track of important tasks. As a result, they may forget deadlines, delay submissions, or miss important academic responsibilities.

The lack of a simple and structured system for task management leads to poor time management and increased stress among students. Many available task management tools are either too complex or not specifically designed for basic student needs.

Therefore, there is a need for a simple, user-friendly application that allows students to efficiently manage their tasks in one place. The system should help users easily add, view, and remove tasks so that they can stay organized and improve their productivity.

Objective

The objectives of this project are:

- To design a simple task management application
  - To help students organize their daily tasks
  - To implement Java programming concepts practically
- To create an easy-to-use system for beginners

Scope of the Project

The scope of this project is to develop a simple console-based application that helps students manage their daily tasks efficiently. The system allows users to perform basic operations such as adding new tasks, viewing existing tasks, and deleting completed tasks.

This application is designed for students who need an easy and lightweight solution for organizing their academic work. It runs locally on a system and does not require internet access, making it simple to use.

The project is limited to a text-based interface and does not include advanced features like data storage or graphical design. However, it provides a basic structure that can be improved in the future by adding features such as deadlines, priority levels, and a graphical user interface.

Methodology

The development of this project follows these steps:

1.Understanding the problem of task management

2.Planning a simple solution using Java

3.Implementing features such as adding, viewing, and deleting tasks

4.Testing the application to ensure it works correctly

Technologies Used

Java (programming language)

ArrayList (for storing tasks dynamically)

Scanner class (for user input)

System Design

Input:

- User enters menu choices
- Task details or task number

Process:

- Tasks are stored in a dynamic list
- Operations are performed based on user input

Output:

- Display list of tasks
- Show confirmation messages

Features

- Add tasks easily
- View all stored tasks
- Delete completed tasks
- Simple menu-based interface

Implementation Details

The system is built using Java and follows a structured approach. A class is used to represent tasks, and an ArrayList is used to store them dynamically. The program runs in a loop to continuously accept user input, and different operations are handled using conditional statements.

CODE:

TaskManager.java

C: > Users > Dell > Desktop > BYOP_Project > TaskManager.java

```java
import java.util.ArrayList;
import java.util.Scanner;

class Task {
    String name;

    Task(String name) {
        this.name = name;
    }
}

public class TaskManager {
    public static void main(String[] args) {
        ArrayList<Task> tasks = new ArrayList<>();
        Scanner sc = new Scanner(System.in);

        while (true) {
            System.out.println("\n--- Student Task Manager ---");
            System.out.println("1. Add Task");
            System.out.println("2. View Tasks");
            System.out.println("3. Delete Task");
            System.out.println("4. Exit");
            System.out.print("Enter choice: ");

            int choice = sc.nextInt();
            sc.nextLine();

            switch (choice) {
                case 1:
                    System.out.print("Enter task name: ");
                    String name = sc.nextLine();
                    tasks.add(new Task(name));
                    System.out.println("Task added!");
                    break;
```

```java
            case 2:
                if (tasks.isEmpty()) {
                    System.out.println("No tasks available.");
                } else {
                    System.out.println("\nYour Tasks:");
                    for (int i = 0; i < tasks.size(); i++) {
                        System.out.println((i + 1) + ". " + tasks.get(i).name);
                    }
                }
                break;

            case 3:
                System.out.print("Enter task number to delete: ");
                int index = sc.nextInt();
                if (index > 0 && index <= tasks.size()) {
                    tasks.remove(index - 1);
                    System.out.println("Task deleted!");
                } else {
                    System.out.println("Invalid task number.");
                }
                break;

            case 4:
                System.out.println("Exiting...");
                return;

            default:
                System.out.println("Invalid choice.");
        }
    }
}
```

OUTPUT:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\Dell> cd "c:\Users\Dell\Desktop\BWOP_Project\" ; if ($?) { javac TaskManager.java } ; if ($?) { java TaskManager }

--- Student Task Manager ---
1. Add Task
2. View Tasks
3. Delete Task
4. Exit
Enter choice: 1
Enter task name: Student name
Task added!

--- Student Task Manager ---
1. Add Task
2. View Tasks
3. Delete Task
4. Exit
Enter choice: 1
Enter task name: Registration number of the student
Task added!

--- Student Task Manager ---
1. Add Task
2. View Tasks
3. Delete Task
4. Exit
Enter choice: 2

Your Tasks:
1. Student name
2. Registration number of the student

--- Student Task Manager ---
1. Add Task
2. View Tasks
3. Delete Task
4. Exit
Enter choice: 3
Enter task number to delete: 2
Task deleted!
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\Dell> cd "c:\Users\Dell\Desktop\BYOP_Project\" ; if ($?) { javac TaskManager.java } ; if ($?) { java TaskManager }
Task deleted!

--- Student Task Manager ---
1. Add Task
2. View Tasks
3. Delete Task
4. Exit
Enter choice: 2

Your Tasks:
1. Student name

--- Student Task Manager ---
1. Add Task
2. View Tasks
3. Delete Task
4. Exit
Enter choice: 1
Enter task name: Proctors details
Task added!

--- Student Task Manager ---
1. Add Task
2. View Tasks
3. Delete Task
4. Exit
Enter choice: 2

Your Tasks:
1. Student name
2. Proctors details

--- Student Task Manager ---
1. Add Task
2. View Tasks
3. Delete Task
4. Exit
Enter choice: 4
Exiting...
PS C:\Users\Dell\Desktop\BYOP_Project>
```

Result

The application works as expected and allows users to manage tasks efficiently. All features such as adding, viewing, and deleting tasks function properly.

Learning Outcomes

- Gained understanding of Java basics
- Learned how to use collections like ArrayList
- Improved logical thinking and problem-solving skills
- Understood how to build a simple real-world application

Future Enhancements

- Adding deadlines for tasks
- Including priority levels
- Developing a graphical user interface
- Connecting to a database for storage

Conclusion

The Student Task Manager project successfully demonstrates how a simple application can help solve a common problem faced by students in managing their daily academic tasks. By providing basic functionalities such as adding, viewing, and deleting tasks, the system helps users stay organized and reduces the chances of forgetting important work.

This project highlights the practical application of Java programming concepts, including the use of collections, object-oriented design, and user input handling. It also shows how even a small-scale application can improve productivity and time management when designed effectively.

Although the current system is simple and console-based, it serves as a strong foundation for future improvements. Additional features such as task deadlines, priority levels, and a graphical user interface can further enhance its usability and effectiveness.

Overall, the project meets its objectives and provides a useful tool for students while also strengthening programming skills and problem-solving abilities.